Polish vs. English: measuring the LLM performance gap on matched NLP tasks
Project Proposal for NLP Course, Winter 2026/27

Piotr Czarnecki
Album number 327335
01180766@pw.edu.pl

Emil Łasocha
Album number 341168
01202252@pw.edu.pl

Patryk Sukiennik
Album number 327254
01180679@pw.edu.pl

Michał Szewczak
Album number 327320
01180750@pw.edu.pl

supervisor: Michał Gromadzki
Warsaw University of Technology
michal.gromadzki.dokt@pw.edu.pl

Template: Anna Wróblewska (Warsaw University of Technology), adapted from the ACL 2015 conference formatting template.

Abstract

Large Language Models are evaluated mostly in English, but users also work in different languages such as Polish. The proposal of the project consists of building a small and reproducible Polish–English benchmark, evaluating several Large Language Models under identical prompts, and quantifying how much of the observed performance gap is due to model size, prompting and translation quality.

1 Introduction

Large language models are typically assessed mainly in English, but their effectiveness in Polish may vary considerably. Examining these differences helps reveal the limitations and weaknesses that multilingual models still face. Our preliminary hypothesis is that LLM performance differs between English and Polish, with a general tendency toward lower performance on Polish inputs. However, the magnitude and direction of this difference may vary across tasks, model families, model sizes, and prompting conditions. The project fixes four tasks, and a controlled grid of models. This allows a decomposition of the gap rather than only its measurement.

We propose the following specific research goals:

- a Polish–English benchmark covering four tasks, with a documented translation procedure.
- results for multiple LLMs under matched conditions.
- quantitative analysis of the performance gap across tasks, model sizes, and model families, with additional investigation of tokenization effects where possible.

Week	Tasks and deliverables	Responsible
1	Dataset inspection, model and subset selection, repository setup, and experiment planning.	All four persons
2	English subset sampling, Polish translation, translation documentation, and exploratory data analysis.	Michał, Patryk, Emil
3	Translation verification, back-translation checks, evaluation pipeline development, and proof of concept with the first result.	Piotr, Patryk
4	Prompt finalization, baseline selection, metric definition, preliminary evaluation, and runtime/memory measurements.	Emil, Michał
5–6	Main experiments across models, languages, and tasks; repeated runs for uncertainty estimation.	Patryk, Piotr, Michał
7	Statistical analysis, performance comparison, tokenization analysis, error analysis, and limitations.	Emil, Piotr
8	Final report, revised literature review, reproducibility checks, README, and code release.	All four persons

Table 1: Project timeline, task allocation, and responsibilities over eight weeks.

The concept work of the project are presented in Table 1.

2 State-of-the-Art Papers

- BoolQ (Clark et al., 2019) evaluates whether a model can answer natural yes/no ques-

tions using a supporting passage. It uses approximately 16.000 naturally occurring questions paired with Wikipedia paragraphs. The model receives a question and its passage and predicts either "yes" or "no." The best reported result is 80.43% accuracy, achieved by BERT-large with MultiNLI pre-training, fine-tuned on BoolQ. This dataset will be used as one of the benchmarks in our project.

- GSM8K (Cobbe et al., 2021) introduces curated dataset of 8.5 thousand high-quality grade school math problems with natural-language solutions. It then proposes to employ training verifiers to improve LMM's multi-step mathematical reasoning. The generator was fine-tuned for 2 epochs. 100 solutions per training problem were then sampled from it. Each solution was labelled as correct or incorrect, based only on the final answer. Then a separate verifier was trained for 1 epoch to rank solutions. Verifiers help achieve better performance than the same LLM with 30x model size and they scale well with additional data.

- MGSM (Shi et al., 2022) evaluates the multilingual mathematical reasoning abilities of Large Language Models. It consists of 250 grade–school math problems from the GSM8K dataset, manually translated into ten diverse languages. The model receives a mathematical question in one of the target languages and generates a numerical answer, either directly or by providing intermediate reasoning steps using chain-of-thought prompting. The best reported result is 55.0% accuracy, achieved by PaLM-540B using English chain-of-thought reasoning after translating the questions into English.

- Okapi (Lai et al., 2023) introduces a framework for instruction-tuning large language models in multiple languages using reinforcement learning from human feedback (RLHF). It covers 26 languages and provides approximately 158.000 instruction examples translated from English. The models are evaluated on their ability to follow instructions and perform knowledge-based and reasoning tasks across different languages. The best reported average result is 49.6% accuracy on HellaSwag, achieved by the LLaMA-7B model fine-tuned using RLHF.

- Self-Translate (Etxaniz et al., 2023) investigates whether multilingual language models perform better on reasoning tasks when processing inputs in English rather than in their original language. It introduces a method called self-translate, in which the model first translates the input into English using four-shot prompting and then solves the task using the translated text. The approach is evaluated on five multilingual benchmarks: XStoryCloze, XCOPA, XNLI, PAWS-X, and MGSM, covering common-sense reasoning, natural language inference, paraphrase identification, and mathematical reasoning across multiple languages. Experiments are conducted using XGLM models ranging from 564M to 7.5B parameters and LLaMA models ranging from 7B to 30B parameters. The best reported average result is 46.6% accuracy, achieved by LLaMA-30B using self-translate, compared to 43.1% with direct inference. The results show that translating inputs into English consistently improves performance across the evaluated models.

- XNLI (Conneau et al., 2018) introduces a cross-lingual natural language inference (NLI) benchmark for evaluating cross-lingual sentence representations. It extends the MultiNLI corpora to 15 languages, including low-resource ones. Consisting of 7500 human-annotated development and test examples in NLI three-way classification format, it makes up a total of 112,500 annotated pairs.

- PLLuM (Kocoń et al., 2025) presents a family of open-source polish foundation models, ranging from 8B to 70B parameters. A 140 billion token polish pretraining corpus was constructed from various texts. Special care was taken to extensively document legal compliance under polish and EU's framework. This model was later used in mObywatel as a virtual assistant.

- Artetxe et al. (Artetxe et al., 2020) shows that human and machine translations can introduce subtle artifacts into data, which cross-lingual models are sensitive to. Fine-tuning

on back-translated (round-trip translated) text improves performance on translated test improves accuracy in the translate-test setting by about 4.6 points on average. Training on machine-translated text also improved results, by about 2 points in zero-shot setting. Improvements only apply to translated texts. Based on these results, XNLI state of the art was improved by about 4.3 points in translate-test and 2.8 points in zero-shot.

- LLMzSzŁ (Jassem et al., 2025) introduces a comprehensive Polish benchmark based on nearly 19,000 multiple-choice questions from official examinations of the Polish Central Examination Board, covering 154 domains and four examination types. The authors evaluate 38 language models using answer likelihood and accuracy to compare multilingual and Polish-specific models. The best result is 67.17% accuracy, achieved by Mistral-Large-Instruct-2407, followed by Llama-3.1-70B-Instruct with 66.59%. The smaller Polish-focused Bielik-11B-v2.1-Instruct achieves 57.52%, demonstrating that language-specific models can remain competitive despite having fewer parameters.

- Bielik 7B (Ociepa et al., 2025) presents a Polish language model with 7 billion parameters, developed through continued pretraining of Mistral 7B. The model was trained on approximately 36 billion Polish and English tokens and fine-tuned using over 2.3 million instruction-response pairs. The authors introduce adaptive learning rate and weighted instruction loss to improve training. In the Open PL LLM Leaderboard's five-shot evaluation, Bielik achieves a RAG Reader score of 88.39, compared to 85.39 for Mistral-7B-v0.1. The results demonstrate that language-specific adaptation can improve performance on Polish question-answering tasks.

- Belebele (Bandarkar et al., 2024) introduces a multilingual reading comprehension benchmark covering 122 language variants, including Polish and English. It contains 900 multiple-choice questions per language, based on 488 passages from FLORES-200, with four possible answers per question. The parallel dataset enables direct comparisons of language model performance across languages. The authors evaluate multilingual and generative language models using zero-shot, and fine-tuning approaches. XLM-V achieves 60.2% average accuracy across all languages using Translate-Train-All, outperforming GPT-3.5 Turbo, which achieves 51.1% in zero-shot evaluation.

- LEPISZCZE (Augustyniak et al., 2022) introduces a comprehensive benchmark for Polish natural language processing, extending previous evaluation frameworks such as KLEJ. It includes 13 task-dataset configurations covering sentiment analysis, named entity recognition, part-of-speech tagging, and text classification. Five Transformer model configurations, including HerBERT, PolBERT, and XLM-RoBERTa, are evaluated using optimized fine-tuning procedures. HerBERT-large achieves the best overall average rank of 1.62, with 96.14% macro-F1 on NKJP-POS and 39.44% on AspectEmo. The results highlight substantial differences in model performance across Polish NLP tasks.

- ARC (AI2 Reasoning Challenge) (Clark et al., 2018) introduces a benchmark designed to evaluate scientific reasoning and question-answering abilities beyond simple information retrieval. It contains 7,787 multiple-choice science questions, divided into 2,590 Challenge and 5,197 Easy questions. The Challenge subset includes 1,172 test questions that cannot be correctly answered by two simple baseline methods. The authors evaluate several reasoning and neural models, including BiDAF and DGEM. The best reported Challenge test result is 27.11% for DGEM, compared to 25.02% for the random baseline, demonstrating the difficulty of questions requiring deeper reasoning. ARC-Challenge will be one of the datasets used in our project.

- Huang et al. (Huang et al., 2023) introduces Cross-Lingual-Thought (XLT) prompting, a strategy that encourages Large Language Models (LLMs) to engage in cross-lingual and logical reasoning by translating or rephrasing prompt queries into English before executing step-by-step reasoning. It is

evaluated across arithmetic reasoning, multi-hop reasoning, and multilingual NLP benchmarks in multiple languages, such as MGSM and Native Reasoning datasets. The approach demonstrates consistent performance gains over direct native-language prompting by bridging cross-lingual reasoning gaps and leveraging English-centric knowledge representation, achieving a $+8.5\%$ average accuracy improvement over direct prompting on MGSM.

- Liu et al. (Liu et al., 2025) investigates the trade-offs and limitations of translation-based prompting versus native-language prompting, evaluating when English translation helps or hinders task performance. The study evaluates models on standard NLP reasoning tasks as well as culture-specific and socio-linguistic benchmarks across high- and low-resource languages. The results show that while translating to English improves performance on standard NLP tasks for English-centric models, native-language prompting significantly outperforms translation on culture-related tasks requiring deep local context, achieving up to 12.4% higher accuracy than English translation.

- Wendler et al. (Wendler et al., 2024) applies mechanistic interpretability techniques, specifically the logit lens, to analyze intermediate layer representations of multilingual transformer models like the Llama-2 family during non-English processing. The study tests models on parallel multilingual text inputs across high-, medium-, and low-resource languages. The analysis maps internal model transitions showing a progression from the input language space to an intermediate, English-biased "concept space," and finally decoding into the target language output space, exhibiting greater than 70% probability alignment with English concepts in intermediate layers (layers 12–24).

- Xuan et al. (Xuan et al., 2025) introduces MMLU-ProX, a massively multilingual benchmark designed for robust evaluation of reasoning and knowledge across typologically diverse languages using parallel, high-quality questions. The benchmark comprises parallel evaluation sets across 29 diverse languages spanning multiple domains and subjects. The evaluation reveals severe performance degradation in state-of-the-art LLMs on low-resource and non-Western languages, achieving an average accuracy of 68.2% on high-resource splits compared to just 31.4% on low-resource splits such as African languages, representing a performance drop of approximately 36.8%.

- Hendrycks et al. (Hendrycks et al., 2021) proposes a massive multitask language understanding test covering 57 tasks, including STEM, humanities, social sciences, law, and ethics, evaluated under zero-shot and few-shot settings to measure a model's extensive world knowledge and problem-solving abilities. The dataset consists of 15,908 total multiple-choice questions collected from online sources, professional practice exams, and university-level materials. The evaluation finds that while the 175B parameter GPT-3 model outperforms random chance with an average accuracy of 43.9%, its performance remains subhuman across all tasks, exhibiting lopsided weaknesses in calculation-heavy and value-laden subjects, along with significant model miscalibration.

- Petrov et al. (Petrov et al., 2023) demonstrates how disparities in the treatment of different languages arise systematically at the tokenization stage prior to model invocation, introducing the concept of "tokenizer parity" and advocating for multilingually fair subword tokenizers. The approach is evaluated across the FLORES-200 parallel corpus spanning 200 languages using English-centric, multilingual, and byte-level tokenizers. The results show that tokenization lengths for identical semantic content can vary by up to 15 times—such as with the ChatGPT/GPT-4 tokenizer (cl100k_base) on low-resource languages like Shan relative to English—directly introducing severe economic, latency, and context-length inequalities.

Dataset	Size	Label definition	Class balance	Licence	How obtained	Status
MMLU	14,042 test questions	Multiple-choice answers (4 options)	Some subjects contain many questions (more than 1500), but most have much less (up to 200)	MIT licence	Public dataset	READY + TRANSLATION TO COLLECT
GSM8K	7,473 train records and 1,319 validation records	Numeric answers	Not applicable	MIT licence	Public dataset	READY + TRANSLATION TO COLLECT
ARC-Challenge	2,590 questions (1,119 train, 299 validation, 1,172 test)	Multiple-choice answers (one correct option)	Not applicable (no semantic class labels)	CC BY-SA 4.0	Public dataset	READY + TRANSLATION TO COLLECT
BoolQ	9,427 train records and 3,270 validation records	Question, Paragraph, Answer	Majority of True label, 62.31%	CC-BY-SA-3.0	Public dataset	READY + TRANSLATION TO COLLECT

Table 2: Overview of the datasets.

3 Benchmarks

As shown in Table 2, we consider four popular English benchmarks for our study. The original English datasets are publicly available – READY, while Polish translations are TO COLLECT. Table 2 also lists the licences under which the benchmarks are distributed. The BoolQ dataset does not contain any personal information, as its answers are based on Wikipedia paragraphs (Clark et al., 2019). The GSM8K dataset is made up of grade school math questions and it does not contain any personal informations (Cobbe et al., 2021). Similarly, the ARC-Challenge and MMLU datasets are test questions & answers, hence they do not contain personal information.

4 Approach and research methodology

Tasks and data

The study uses subsets of four benchmarks: MMLU, GSM8K, ARC-Challenge and BoolQ. Fixed random samples are selected (initially estimating 500) from their official test splits, with stratification by subject for MMLU. English samples are translated into Polish while preserving the original dataset structure.

Translation procedure

Machine translation is used to create one Polish version of each selected dataset subset. Automatic checks verify the preservation of answer options, numbers, units, and text completeness. A randomly selected sample is reviewed by team members. Translation settings and decisions are documented. Potential translation errors are considered when interpreting the results and are discussed as a limitation of the study.

Models and prompting

The evaluation covers instruction-tuned open-weight models from multiple families, including different model sizes and a Polish-focused model such as Bielik. A commercial model may also be included.

Models are evaluated using zero-shot prompting. Answers are evaluated using exact match. We consider two prompt conditions: English data with an English instruction, and Polish data with a Polish instruction. The question is identical across conditions; only the language of the question changes. Decoding is greedy, so results are deterministic; uncertainty is estimated with paired bootstrap confidence intervals over items.

Metrics and analysis

The main metric is the accuracy gap between English and Polish:

$$\Delta = \mathrm{Acc}_{EN} - \mathrm{Acc}_{PL}.$$

Results are analysed using 95% confidence intervals and statistical significance tests. To decompose the gap, we compare Δ across tasks, across model families, and across model sizes within the same family, which shows how much of the gap depends on each factor. To estimate the effect of translation quality, we compute Δ separately on items that passed manual and back-translation

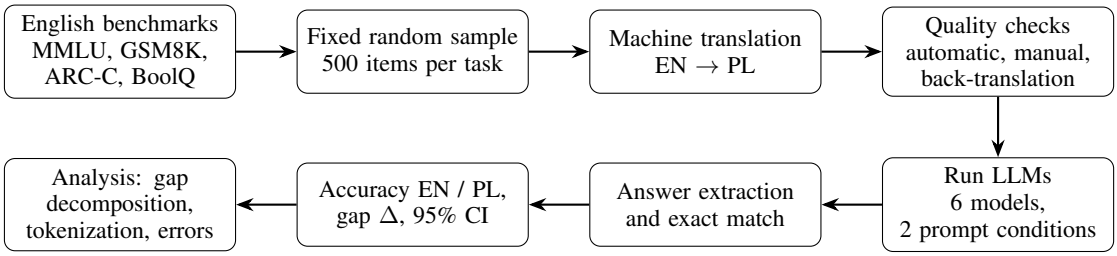


Figure 1: Experimental pipeline. The same 500 items per task are evaluated in English and in their Polish translation, so every comparison is paired.

review and on items flagged as problematic; the part of the gap that remains on the clean items is attributed to language-specific difficulty. Additional analyses examine differences across tasks and model sizes, and the relationship between tokenization and model performance, for models supporting tokenizer access. We compare the accuracies to a baseline model in order to compare the accuracy results with a random guess. Potential effects of translation errors are considered when interpreting the results. A manual error analysis investigates cases where models answer correctly in English but incorrectly in Polish.

Solution concept and success deliverables

Figure 1 shows the pipeline. Metrics and analyses are fixed before any results are available. Success deliverables:

1. A released benchmark of 4 tasks $\times$ 500 items in English and Polish, with documented and verified translation.
2. Evaluation of 6 instruction-tuned LLMs under matched conditions.
3. A quantitative decomposition of the gap with confidence intervals, and error analyses.

5 Contribution table and declaration of LLM use

The contribution for the project proposal are as follows:

- Piotr: $\approx$ 5 hours, describing the 5 papers and the GSM8K dataset
- Emil: $\approx$ 8 hours, describing the 4 papers and BoolQ Dataset, the introduction, methodology and project timetable paragraph.
- Patryk: $\approx$ 4 hours, describing the 5 papers and ARC Dataset
- Michał: $\approx$ 5 hours, describing the 5 papers and the MMLU dataset

The use of LLMs is limited to the following tasks:

- Obtaining an initial overview of the project.
- Correcting grammatical errors.
- Reviewing the text for substantial issues and inconsistencies.
- Generating LaTeX tables.
- Drafting short passages, followed by verification and manual revision of the generated content.

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